



UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE
SEMESTER I - 2021/2022



MAT1073: Mathematics for Operations Research
End Semester Examination

Time allowed: THREE Hours

Answer All Questions

1. (a) Let $V = \{(x_1, x_2) \mid x_1, x_2 \in \mathbb{R}\}$. Determine whether set V forms a vector space over $\mathbb{R}$ under the given addition and multiplication.
 - i. **Addition:** $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$
Multiplication: $\alpha(x_1, x_2) = (\alpha x_1, \alpha x_2)$, where $\alpha \in \mathbb{R}$
 - ii. **Addition:** $(x_1, x_2) + (y_1, y_2) = (x_1 + y_1, x_2 + y_2)$
Multiplication: $\alpha(x_1, x_2) = (2\alpha x_1, 2\alpha x_2)$, where $\alpha \in \mathbb{R}$
- (b) Given a subset W of a vector space V over $\mathbb{R}$. Determine whether W is a subspace of V in each case.
 - i. Let V be the set of $n \times n$ matrices with real coefficients, and $W = \{v \in V \mid vw = wv \text{ for all } w \in V\}$.
 - ii. Let $V = \mathbb{R}^3$ and $W = \{(x, 12, 2x) \mid x \in \mathbb{R}\}$.
- (c) Define the term *basis* of a vector space.
Prove that the set $\left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 5 \end{pmatrix} \right\}$ is a basis for $\mathbb{R}^3$.

(100 marks)

2. (a) Consider the matrix

$$M = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 0 & -4 \\ -1 & 5 & 7 \end{pmatrix}$$

- i. Show that matrix M is non-singular.
 - ii. Find the adjoint matrix of M .
 - iii. Find the inverse of M .
- (b) Find the eigenvalues and the corresponding eigenvectors of the matrix A ;

$$A = \begin{pmatrix} 0 & 3 & -2 \\ -2 & 5 & -2 \\ -2 & 3 & 0 \end{pmatrix}$$

- (c) Solve the following system of linear equations using Cramer's rule.

$$\begin{aligned} x + y + z &= 6 \\ y + 3z &= 11 \\ x - 2y + z &= 0 \end{aligned}$$

(100 marks)

3. (a) Salvador's Fruit Mart sells variety packs. The small pack contains three bananas, two apples, and one orange for 1.80 dollars. The medium pack contains four bananas, three apples, and two oranges for 3.05 dollars. The family pack contains bananas twice as many as in a small pack, five apples, and four oranges for 4.65 dollars.

Formulate a mathematical model to find Salvador's charge for a special pack that contains one of each fruit. Solve the formulated model by using Gaussian Elimination Method.

- (b) Express the following system of linear equations in the form $A\underline{x} = B$ where A , $\underline{x}$ and B are matrices in appropriate dimensions.

$$\begin{aligned} x_1 + 3x_2 + 2x_3 &= 2 \\ 2x_1 + 2x_2 + 3x_3 &= 5 \\ 3x_1 + 4x_2 + 5x_3 &= 8 \end{aligned}$$

Using LU-decomposition of matrix A , solve the above system of linear equations.

(100 marks)

4. (a) Show that the following differential equation is exact, and hence solve it:

$$x(1 - \sin y) \frac{dy}{dx} + y = \cos x - \cos y$$

- (b) Solve the following differential equations.

i. $\frac{dy}{dx} = \frac{3y - 7x + 7}{7y - 3x + 3}$

ii. $\frac{dy}{dx} = 3y + x^5 y^{\frac{1}{3}}$

- (c) Find the general solution of each of the following differential equations.

i. $(D - 3)^8(D - 5)^2 y = 40e^{5x}$

ii. $(D^3 - 5D^2 - D + 5)y = 29 \sin 2x$

(100 marks)

.END.